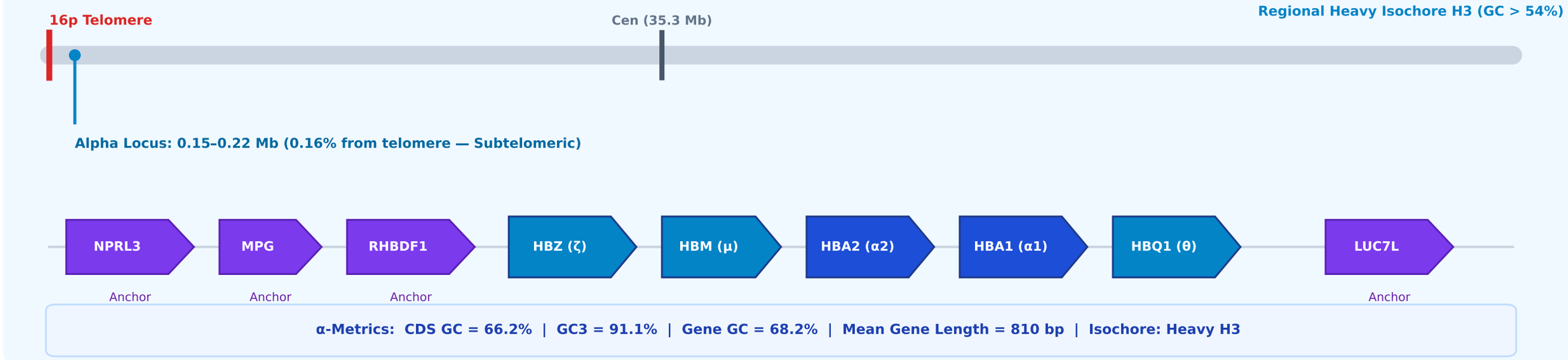
A. The Human Globin Paradigm & The Core Evolutionary Inquiry

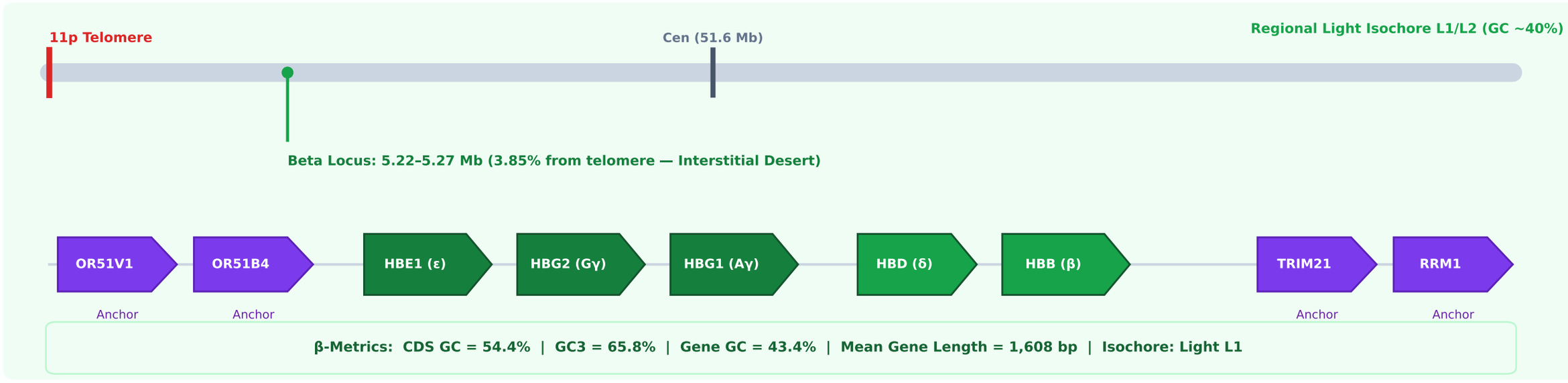
Canonical dual-cluster arrangement on chromosomes 16 and 11, and the hypothesis of universal isochore polarity

Human Alpha-Globin Cluster — Subtelomeric Heavy Isochore (16p13.3)



α-Metrics: CDS GC = 66.2% | GC3 = 91.1% | Gene GC = 68.2% | Mean Gene Length = 810 bp | Isochore: Heavy H3

Human Beta-Globin Cluster — Interstitial Light Isochore (11p15.4)



β-Metrics: CDS GC = 54.4% | GC3 = 65.8% | Gene GC = 43.4% | Mean Gene Length = 1,608 bp | Isochore: Light L1

The Core Macroevolutionary Inquiry:

1. Is this extreme GC disparity (subtelomeric high-GC α vs. interstitial low-GC β) an invariant vertebrate feature?
2. Did physical distance to the telomere dictate locus GC content via GC-biased gene conversion (gBGC) across 500 Myr?